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$$\begin{aligned} & \int \frac{4x + 3\sqrt{x^2 - 1}}{x\sqrt{x^2 - 1}} dx \\ &= \int \frac{4x}{x\sqrt{x^2 - 1}} dx + \int \frac{3\sqrt{x^2 - 1}}{x\sqrt{x^2 - 1}} dx \\ &= 4 \int \frac{1}{\sqrt{x^2 - 1}} dx + 3 \int \frac{1}{x} dx \\ &= 4 \ln |x + \sqrt{x^2 - 1}| + 3 \ln |x| + C \end{aligned}$$

References